Syn Bank Share of Wallet Intelligence Engine

Standard Bank Data School Hackathon 2026 • Team ROOT3 • Luke Naidoo · Wisdom Ejiro Peru · Fatan Saud

20	5.8%	R2.4tn	0.24–0.39	3/3
JSE-listed clients analyzed (43 company-years)	avg. blended Share of Wallet, actionable-tier clients	combined addressable gap, 10 highest-confidence clients	ElasticNet R^2, 4/5 targets (honest LOGO cross-validation)	GenAI use cases built: briefing notes, NL query, anomalies

THE PROBLEM

Syn Bank is never a client's **sole** banking partner. We built an engine that estimates each client's **Total Wallet** across three product pillars, quantifies Syn Bank's **current Share** from internal data, and ranks the **Rand Gap** to prioritize which clients a coverage banker should chase next — for the 20 JSE-listed corporates in Syn Bank's synthetic transactional, trade finance, and cross-border datasets.

OUR APPROACH

Numerator (Share): internal transactional, trade finance and cross-border data, split into 3 pillars — Transactional Banking, Trade & Working Capital, Foreign/Cross-Border.

Denominator (Wallet): top-down proxy built from each client's own disclosed annual-report financials (revenue, cost of sales, receivables/payables, foreign-revenue %), extracted from 43 company-year filings (FY2024–2026) and normalized to ZAR at live SARB rates.

Learned angle: ElasticNet regression (5 targets, Leave-One-Group-Out CV) predicts Share % directly from internal activity volume — useful precisely where the top-down external benchmark is thin or missing.

GENAI LAYER (3/3 USE CASES)

Client briefing notes — all 20/20 clients, machine-verified against source data (142/143 numeric claims auto-confirmed; caught one real stale-data error before it shipped).

Anomaly explanations — code detects (60 client×pillar cells, fixed rules), GenAI explains only what code already found (27 anomalies, 6 patterns) — hallucination-resistant by design.

NL query assistant — live, zero-cost, deterministic layer for common lookups; LLM reserved only for genuinely open-ended synthesis questions.

KEY FINDINGS

- **Top opportunity:** Shoprite Holdings — 2.4% transactional share, R437bn gap on a R448bn wallet (literal, ZAR-reporter).
- **Sharpest anomaly:** Valterra Platinum — near-zero share (0.0–0.2%) despite being a *domestic* reporter, not a scale artifact — flagged for immediate relationship review.
- **Model self-awareness as a finding:** reliability tiering catches the naive-model trap before it reaches a banker — a raw calculation would report Glencore's global consolidated revenue as a fictional R9.5 trillion "gap"; explicitly flagged low-reliability/directional-only instead.

LIMITATIONS, STATED PLAINLY

- **Stated scope cut:** top-down financials-based Total Wallet only — bottom-up competitor evidence (SARB BA900, JSE SENS) was descoped for the solo/time-constrained build; documented as the top next step, not hidden.
- Foreign/Cross-Border wallet is estimated for only 4/20 clients (numeric foreign-revenue disclosure required) — flagged "external estimate unavailable" for the rest, never silently zeroed.

NEXT STEPS

- Layer in bottom-up competitor evidence (SENS facility announcements, BA900) to tighten/validate the top-down proxy.
- Extend Pillar 3 (cross-border) external benchmark coverage beyond the current 4/20 clients.
- Automate the GenAI briefing-note pipeline via API to scale from 20 to the full 43-row multi-year panel.

Reliability tiers, stated on every figure:	**10** moderate (ZAR reporter, literal Rand gap)	**9** low (foreign currency, directional % only)	**1** insufficient (Group AFS not disclosed)

Code & full notebook: github.com/wiz-20/Root3 (private — access on request)

Team ROOT3 | Submission: 16 Aug 2026